RAJALAKSHMI ENGINEERING COLLEGE

Nevash S 251001162-2025-IT

N2

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|---|---|
| **Started on** | Thursday, 17 September 2026, 10:12 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 10:15 AM |
| **Time taken** | 2 mins 29 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

**Output Format**

First Line Contains Integer – Floor value for x

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int a[], int low, int high, int x) {
    if (low > high)
        return -1;

    if (x < a[low])
        return -1;

    if (x >= a[high])
        return a[high];

    int mid = (low + high) / 2;

    if (a[mid] == x)
        return a[mid];

    if (a[mid] > x)
        return findFloor(a, low, mid - 1, x);

    int result = findFloor(a, mid + 1, high, x);

    if (result == -1)
        return a[mid];

    return result;
}

int main() {
    int n, x, i;

    scanf("%d", &n);

    int a[n];

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    printf("%d", findFloor(a, 0, n - 1, x));

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6<br>1<br>2<br>8<br>10<br>12<br>19<br>5 | 2 | 2 | ✔ |
| ✔ | 5<br>10<br>22<br>85<br>108<br>129<br>100 | 85 | 85 | ✔ |
| ✔ | 7<br>3<br>5<br>7<br>9<br>11<br>13<br>15<br>10 | 9 | 9 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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